

L3 – 8.1/8.2 Sum/Difference and Product/Quotient of Functions

MHF4U

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Part 1: Sum and Difference of Functions

When two functions $f(x)$ and $g(x)$ are combined to form the function $(f + g)(x)$ or $(f - g)(x)$, the new function is called the sum or difference of f and g .

The graph of $f + g$ or $f - g$ can be obtained by adding or subtracting corresponding y-coordinates. This is called the **superposition principle**.

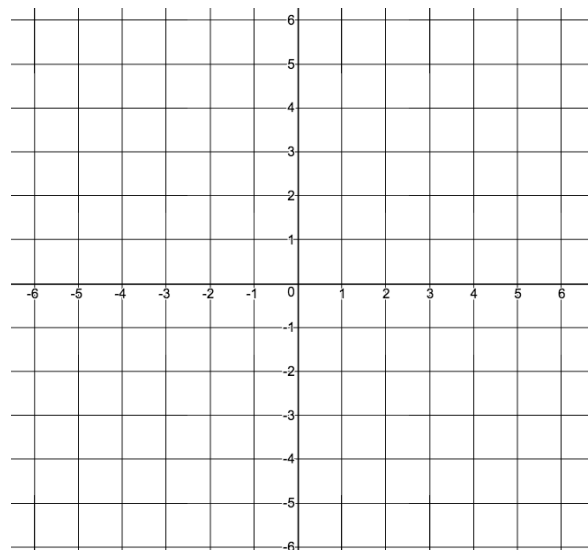
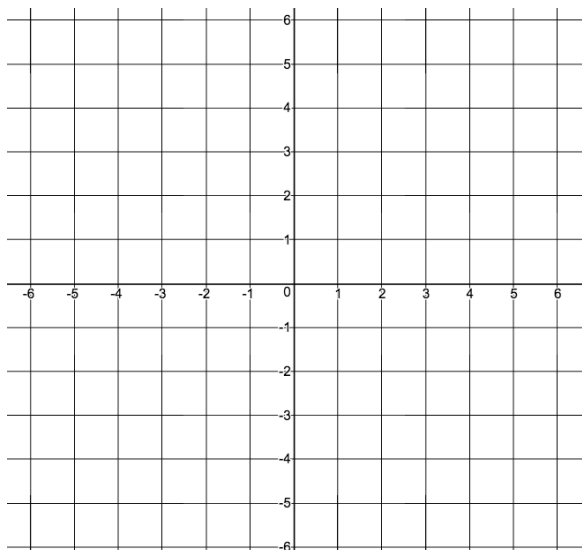
$$(f + g)(x) = f(x) + g(x)$$

$$(f - g)(x) = f(x) - g(x)$$

Example 1: Given $f(x) = -x^2 + 3$ and $g(x) = -2x$ determine the graphs of $(f + g)(x)$ and $(f - g)(x)$.

Method 1: Graphically

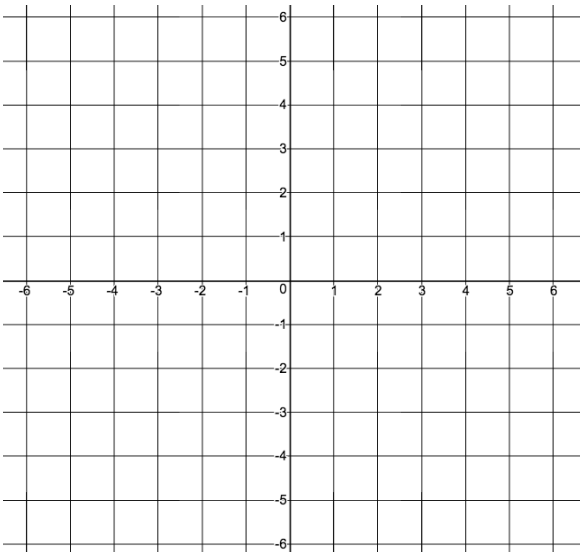
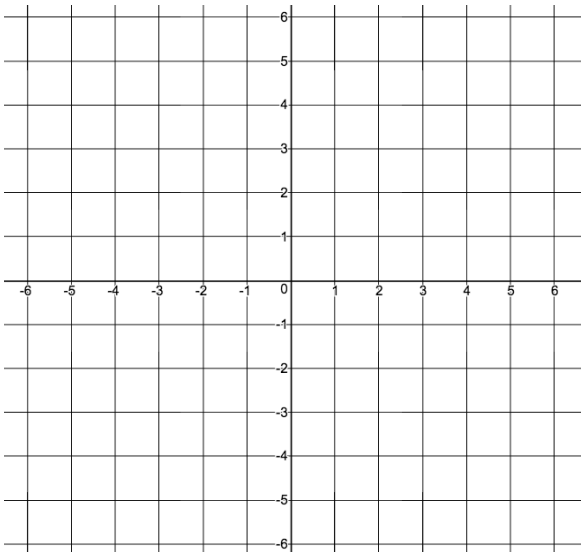
x	$f(x)$	$g(x)$	$f(x) + g(x)$	$f(x) - g(x)$
-3				
-2				
-1				
0				
1				
2				
3				



Method 2: Algebraically

x	$(f + g)(x)$

x	$(f - g)(x)$



Note: The domain of the sum or difference of functions is the intersection of the domains of f and g

Part 2: Product and Quotient of Functions

When two functions $f(x)$ and $g(x)$ are combined to form the function $(f \cdot g)(x)$ or $(f \div g)(x)$, the new function is called the product or quotient of f and g .

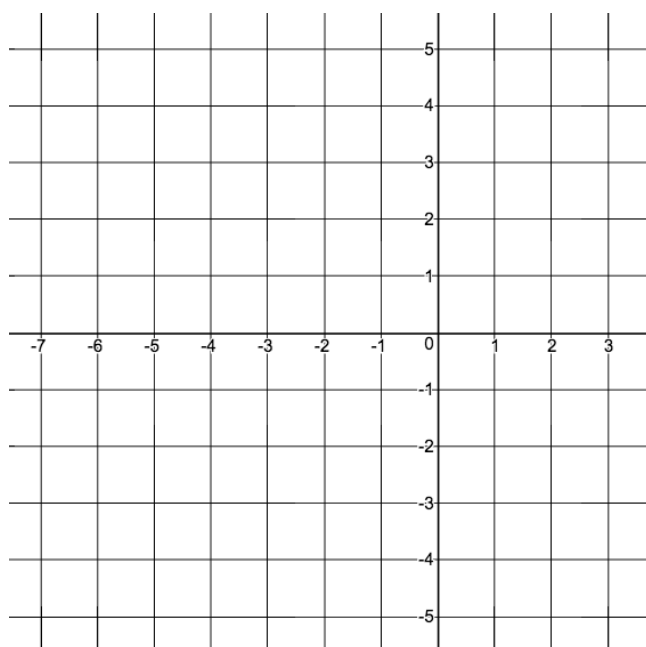
The graph of $f \cdot g$ or $f \div g$ can be obtained by multiplying or dividing corresponding y -coordinates.

$$(f \times g)(x) = f(x) \times g(x)$$

$$(f \div g)(x) = f(x) \div g(x)$$

Example 2: Let $f(x) = x + 3$ and $g(x) = x^2 + 8x + 15$. Determine an equation and graph for

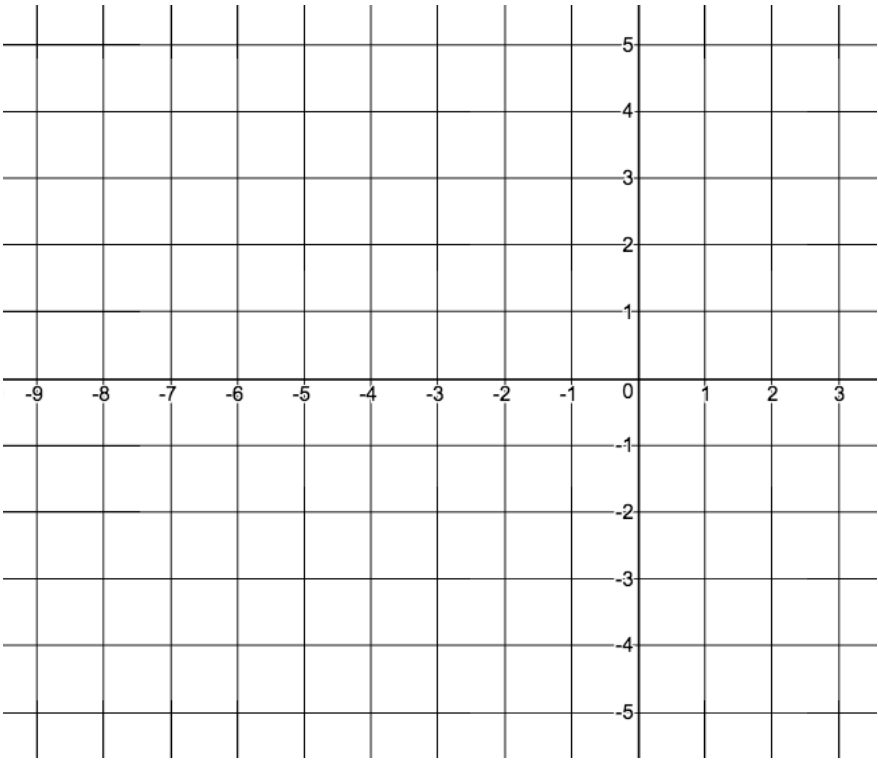
a) $(f \times g)(x)$



b) $(f \div g)(x)$

x	y

x	$\frac{1}{y}$



c) State the domain and range of both functions